Logbook

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- **Week one: Grade 10 Summer camp & the challenge was not yet announced**

Week Two

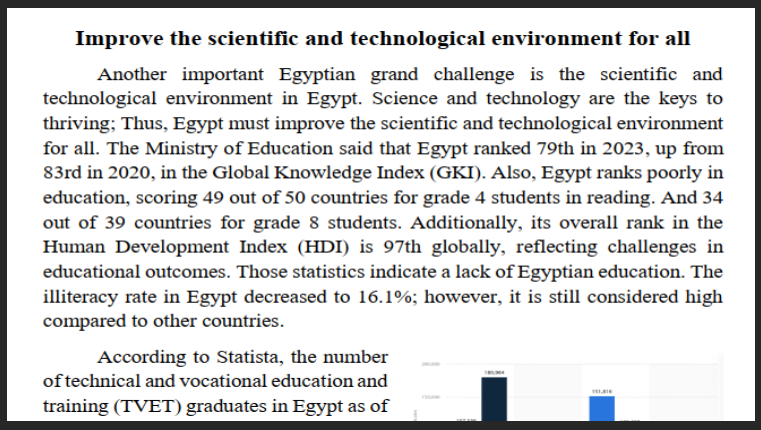
- Brainstorming and building teams in addition to studying the grand challenges.

Week Three

-In the third week my teammates and I began working on the project through extensive research to select the idea based on the just announced challenge-

- Studied the capstone challenge & identified the feasibility of constructing a biological incubator.

- Began working on the first grand challenge, part of which is shown in figure 1.



Improve the scientific and technological environment for all

Another important Egyptian grand challenge is the scientific and technological environment in Egypt. Science and technology are the keys to thriving; Thus, Egypt must improve the scientific and technological environment for all. The Ministry of Education said that Egypt ranked 79th in 2023, up from 83rd in 2020, in the Global Knowledge Index (GKI). Also, Egypt ranks poorly in education, scoring 49 out of 50 countries for grade 4 students in reading. And 34 out of 39 countries for grade 8 students. Additionally, its overall rank in the Human Development Index (HDI) is 97th globally, reflecting challenges in educational outcomes. Those statistics indicate a lack of Egyptian education. The illiteracy rate in Egypt decreased to 16.1%; however, it is still considered high compared to other countries.

According to Statista, the number of technical and vocational education and training (TVET) graduates in Egypt as of

Figure 1, Grand Challenge Sample

- Researched different heat mechanisms for the incubator.

  Sources:

  https://www.sciencedirect.com/science/article/abs/pii/S1350453306001627

  https://ggsceramic.com/news-item/updated-ceramic-heating-elements-all-you-need-to-know

  https://www.ptcyidu.com/application/ceramic-heater-for-incubator

Week Four

-In the fourth week, my teammates and I began identifying the problem to be solved, which I was assigned to write, as the grand challenges were steadily coming out one by one from all the team members-

- Finished writing the grand challenge.

- Began work on the problem to be solved, part of which is shown in figure 2

Problem to be solved

Increasing the capabilities of research in Egypt's laboratories through recycling electronic waste

Electronic waste, defined as any electronic appliance that is considered to have reached the end of its usefulness, has been rapidly increasing in the Egyptian landfill sites & markets. According to GEWSP (The Global E-Waste Statistics Partnership) electronic waste in the Egyptian market (EEE) has increased from 754.0 kt in 2015 to 1069.0 kt in 2019. As e waste generated increased from 477.0 kt in 2015 to 586.0 kt in 2019 & e waste generated (kg per capita) increased from 5.16 kg to 5.85 kg. Additionally, said increase in electronic waste **is reflected in figure 1** through Egypt's rank in electronic waste generation among African countries.

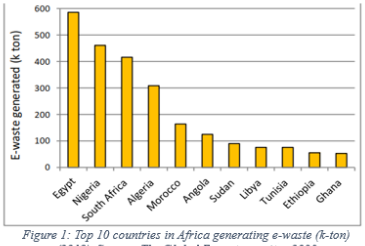


Figure 1: Top 10 countries in Africa generating e-waste (k-ton) (2019). Source: The Global E-waste monitor 2020

Figure 2, Problem to be Solved

Week Five

- Finished writing the problem to be solved.

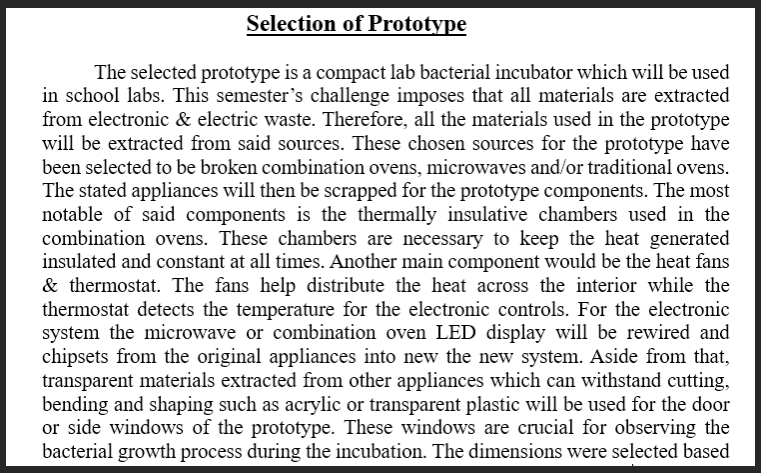
Sources:
https://cen.acs.org/environment/recycling/Electronic-waste-gold-mine-waiting/102/i23
https://www.researchgate.net/publication/326226096_Economic_Feasibility_Study_of_E-Waste_Recycling_Facility_in_Egypt

https://www.researchgate.net/publication/342783104_The_Global_E

Researched electronic thermostats & their operational length.

Week Six

- Wrote the selection of prototype, part of which is shown in figure 3.



Selection of Prototype

The selected prototype is a compact lab bacterial incubator which will be used in school labs. This semester's challenge imposes that all materials are extracted from electronic & electric waste. Therefore, all the materials used in the prototype will be extracted from said sources. These chosen sources for the prototype have been selected to be broken combination ovens, microwaves and/or traditional ovens. The stated appliances will then be scrapped for the prototype components. The most notable of said components is the thermally insulative chambers used in the combination ovens. These chambers are necessary to keep the heat generated insulated and constant at all times. Another main component would be the heat fans & thermostat. The fans help distribute the heat across the interior while the thermostat detects the temperature for the electronic controls. For the electronic system the microwave or combination oven LED display will be rewired and chipsets from the original appliances into new the new system. Aside from that, transparent materials extracted from other appliances which can withstand cutting, bending and shaping such as acrylic or transparent plastic will be used for the door or side windows of the prototype. These windows are crucial for observing the bacterial growth process during the incubation. The dimensions were selected based

Figure 3, Selection of Prototype Sample

Sources:
https://uomus.edu.iq/img/lectures21/WameedMUCLecture_2021_92853824.pdf

- Researched the Peltier effect, and its related cooling systems.

Week Seven

- Researched heat convection systems & ceramic heaters for the construction of the prototype.

- Researched different types of thermal insulators for the prototype.

- Finalized the shape & schematics of the prototype with the help of my teammate.

Week Eight

- Began searching for vendors to purchase the materials.

- Researched electric circuits and power supplies for the upcoming construction of the prototype.

- Sources:
https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.britannica.com/technology/electric-circuit
https://www.bbc.co.uk/bitesize/articles/zjm8kty

Week 9
Midterm
Minimal prototype research

Week 9
Midterm
Minimal prototype research

Week 9
Midterm
Minimal prototype research

Week 10

- Purchased the materials and machines for component extraction, some of **which are shown in figure 4.**



Figure 4, a variety of extracted components

- Completed the 3D design of the prototype **which is shown in figure 5**.



Figure 5, 3D Design

- Researched the exact components extracted and their function.

Week 11

- Began planning the assembly process and drawing the schematic of the circuit **as shown in figure 6**

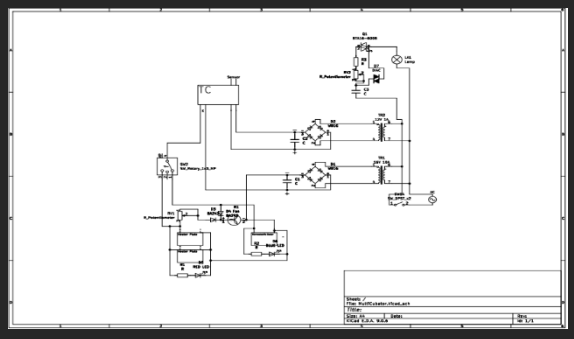


Figure 6 Prototype Circuit Schematic

- Prepared the frame of the prototype from a discarded oven **as shown in figure 7.**



Figure 7 Discarded Oven Frame (before preparation)

- Assembled the panel, electric circuit and components into the frame.

- Completed prototype construction and made sure all the required functions are present.

- Completed the test plan video and results.

- Tested the prototype for errors and failures.

Week 12

- Modified the electric circuit
- Modified the design of the poster, correcting errors and size.
- Filled the poster with its respective content, modifying the materials, methods and test plan content to better present our project.
- Corrected mistakes in the A4 Poster such as graphs and figures.
- Submitted the poster, portfolio, and A4 poster with my team. The final poster **is shown in figure 8.**

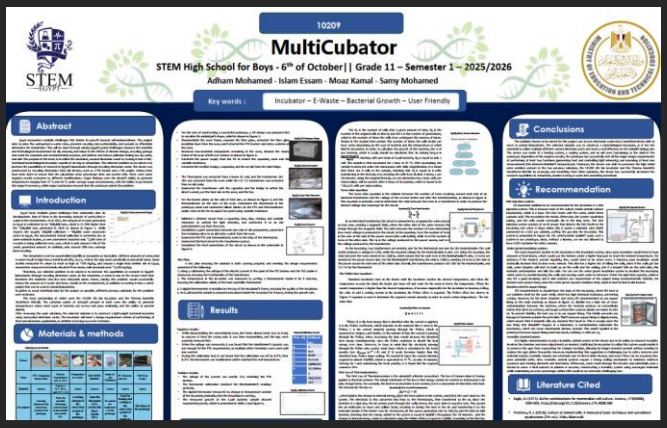


Figure 8 The MultiCubator Final Poster

Week 13

- Prepared for the exhibition through rehearsing the poster and prototype details with the team members

- Conducted trial exhibitions with my team in various areas with different questions and how we can answer them.

Overall, this week was full of preparation and rehearsal.